XRPL Validator — Daily Engineering Report · 2026-09-22

Period: 2026-09-22 00:00 to 24:00 US Eastern (2026-09-22 04:00 to 2026-09-23 04:00 UTC); ledgers #107,151,005 to #107,173,371. Every count from the ledger and the validator's logs stops at #107,173,371. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Degraded — coverage outside planned maintenance 99.802% (objective 99.9%): 44 ledgers unsigned.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,128 of the 22,367 ledgers the network closed (98.931%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 195 during planned restarts, 44 outside maintenance (1 incident).
- **Native engine:** 2 engine receipts recorded (2 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 2 deployments (cycles 149, 150), 4 commits merged into main, 26 engine findings fixed (F357–F382).

Service levels

Measure	Value
Ledgers closed by the network	22,367
Validations issued	22,128
Coverage (validations / ledgers closed)	98.931%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	195
Unsigned outside maintenance	44
Coverage outside planned maintenance (objective 99.9%)	99.802%
Restarts	2

Measure	Value
Basis: the hash the first peer announced as accepted (before cycle 183)	22,128

Restarts:

- 13:25:06 UTC: deployment of cycle 149; validations resumed at #107,159,890 (94 ledgers unsigned between the last validation before the restart and the first after it)
- 15:10:39 UTC: deployment of cycle 150; validations resumed at #107,161,529 (101 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,151,005–#107,173,371), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,155,576	Step 1 · native engine	1 transaction result(s) differed from rippled's	fixed: F357, commit 71fd2d5, deployed in cycle 149 (2026-09-22 13:25 UTC)
R-02	—	#107,159,929	Step 1 · native engine	1 object(s) differed in content	fixed: F358, commit 7c9fde0, deployed in cycle 150 (2026-09-22 15:10 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-22-01, INC-01 is INC-2026-09-22-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

INC-01 · 44 ledgers unsigned

Window	not held: the reference node no longer holds these ledgers' close times
Ledgers	#107,155,086–#107,155,129 (44)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
149	`71fd2d5`	13:25:06 UTC	n/a	n/a	409 matched, 0 mismatched	#24 from 13:52 UTC: ended early: superseded by the next deployment (2026-09-22 15:38 UTC)
150	`7c9fde0`	15:10:39 UTC	n/a	n/a	415 matched, 0 mismatched	#25 from 15:38 UTC: ended early: superseded by the next deployment (2026-09-23 19:57 UTC)

Cycle 149 delivered: `71fd2d5` fix(ledger): F357 — OfferCreate judges "crossed" by the after-cross amounts, not by a dust output

Cycle 150 delivered: `7c9fde0` fix(ledger): F358 — OracleSet rebuilds the series as rippled does: unnamed pairs keep only their asset names, output in token-pair order

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
11:27 UTC	xrpl-validator	`71fd2d5`	fix(ledger): F357 — OfferCreate judges "crossed" by the after-cross amounts, not by a dust output
14:11 UTC	xrpl-validator	`7c9fde0`	fix(ledger): F358 — OracleSet rebuilds the series as rippled does: unnamed pairs keep only their asset names, output in token-pair order
02:19 UTC	xrpl-validator	`c3c915c`	feat(node): receipt canary — the live shadow plants a 1-drop diff on demand and proves its compare still flags it
03:06 UTC	xrpl-sdk	`c6fd55d`	fix(core): encode rippled's "1" spelling of noCurrency (finding 369)

Findings fixed

- **fix(ledger): F357 — OfferCreate judges "crossed" by the after-cross amounts, not by a dust output** (`71fd2d5`, xrpl-validator) Soak #23 receipt (#107155576 E70334ED9A24): rKVBZwTW buys 175 XLM for 37.301075 USD holding 0.958 XRP against a 1.6 XRP reserve. flowCross moved nothing the ledger could see but reported a dust output; rippled's `crossed` is `takerAmount != placeOffer` (OfferCreate.cpp:794) — the amounts were unchanged, so nothing crossed and the reserve gate claimed tecINSUF_RESERVE_OFFER. We set crossed from `actual_out > 0`, took …

- **fix(ledger): F358 — OracleSet rebuilds the series as rippled does: unnamed pairs keep only their asset names, output in token-pair order** (`7c9fde0`, xrpl-validator) Soak #24 receipt (#107159929 B88A96E280D3): raQWZPjc updates 9 of its oracle's 10 pairs. rippled's OracleSet::doApply collects every pair the oracle holds as BaseAsset + QuoteAsset only, then applies the transaction's entries — delete (no AssetPrice), update (AssetPrice; Scale only when sent), add — and emits the std::map order of (BaseAsset, QuoteAsset). The unnamed pair (CLEARPOOL_AUM) therefore loses its …
- **fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0** (`12a043f`, xrpl-validator) fixNFTokenPageLinks and fixCleanup3_2_0 are enabled on mainnet, so any mainnet LedgerStateFix would have applied as tecUNSUPPORTED against the network's code — a guaranteed receipt. Found by checking the 3.4.0 transaction list against dispatch.rs.
- **fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own** (`0e3f28b`, xrpl-validator) PermissionDelegationV1_1 reached majority on mainnet 2026-09-21 (active ~10-05); the engine had no DelegateSet and no delegated fee path. rippled 3.4.0: - Transactor::apply consumes the ACCOUNT's sequence/ticket but payFee charges the fee payer = getInitiator() = sfDelegate when present. apply_common now does the same (TxFields::delegate()). - preFeeBalance_ is the account's balance before the fee. Every site that …
- **fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)** (`28aec17`, xrpl-validator) Campaign 15 (testnet, 138 landed txs: AccountSet depth, SetRegularKey, SignerListSet, 13 multi-signed) — 130/138 byte-exact on the live engine; the 8 misses were five rules: - F361: an unknown SetFlag/ClearFlag (32, 99) is a silent no-op; the `>= 32 → temMALFORMED` rule had no specimen behind it. - F362: the AccountTxnID prior check reads an absent field as the zero hash (getFieldH256 on a non-present optional), not …
- **fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19)** (`0a3dfb4`, xrpl-validator) Campaign 19 (testnet, 114 landed: NFTokenModify, NFTokenCancelOffer, Oracle, DID depth) — 10 misses, four rules: - F366: OracleSet's ±300 s LastUpdateTime window is centred on the header closeTime of the ledger being BUILT = parent close + closeTimeResolution, not the parent close every hasExpired test uses (3-25 tec / 3-26 tes). - F367: OracleSet tests the reserve for every adjustment (+2..-1) against …
- **fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18)** (`fa5735e`, xrpl-validator) Campaign 18 (testnet, 124 landed MPT txs) — 15 misses, four rules, plus one Batch-era rule found reading the same code: - F370: MPT direct payments with a TransferFee. rippled's "No rounding" means no rounding MODE: the cost is multiply(amount, rate) = Number product (16-digit mantissa, to nearest) cast to int64 (to nearest, ties even); a partial delivery is divide(SendMax, rate) = muldiv(num, 1e17, den) + 5 then to …
- **fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)** (`e2c806b`, xrpl-validator) Campaign 20 (testnet, 117 landed: DepositAuth by credentials across Payment, EscrowFinish, PaymentChannelClaim, CheckCash, AccountDelete) — 11 misses, six rules: - F375: the MPT payment arm runs the whole verifyDepositPreauth (expired cleanup, credential-set authorization), not only the by-account preauth. - F376: on a direct XRP payment verifyDepositPreauth — cleanup included — runs only when the amount or the …
- **fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)** (`081001c`, xrpl-validator) Campaign 17 (testnet, 164 landed: the OfferCreate flag matrix — Passive, IoC, FoK (fixFillOrKill), Sell, TickSize, TransferRate, self-cross, expiry, OfferSequence, unfunded/frozen makers, reserve edges, freezes) —

156/164 on the live engine; 7 misses were one rule, in two places: - checkGlobalFrozen on the TakerPays asset, then TakerGets, right after the account and ahead of funding and expiry: an issuer under …

- **fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)** (`9010547` , xrpl-validator) rippled's Batch::preflightSigValidated builds the required batch signers from rb.getInitiator() — the inner's sfDelegate when it carries one, because the delegate signed it — plus an sfCounterparty and a co-signing sfSponsor, each unless it is the outer account (Batch.cpp:423-445). We used the inner's Account, so a validated batch with a delegated inner would be refused here as temBAD_SIGNER. Reachable once …

Other engineering changes

- **feat(node): receipt canary — the live shadow plants a 1-drop diff on demand and proves its compare still flags it** (`c3c915c` , xrpl-validator) A zero-receipt soak only means "zero divergences" if the shadow was comparing and its compare could see a divergence. While [local path] exists (XRPL_NATIVE_SHADOW_CANARY overrides), the next ledger that wrote an AccountRoot the FFI leg agreed on gets one drop added to it in the mirror just before the real compare. The compare must flag it; the result is written as its own …
- **fix(core): encode rippled's "1" spelling of noCurrency (finding 369)** (`c6fd55d` , xrpl-sdk) rippled's to_string(Currency) renders 0x00…01 (noCurrency) as "1"; an Oracle BaseAsset/QuoteAsset can carry it, and the shadow reads rippled JSON, so the object it writes could not be encoded (InvalidCurrency). Found by xrpl-validator campaign 19 (testnet 856F2F37D316 / 0717E2C38424).

Tests and campaigns

- **test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set** (`dd90d91` , xrpl-validator)

Independence program

No independence record for UTC day 2026-09-22; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC
PermissionDelegationV1_1	2026-09-21 11:18 UTC	2026-10-05 11:18 UTC

Amendments enabled on mainnet: 94.

Open items

- INC-01 (#107,155,086–#107,155,129 (44)): cause not recorded.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Time-Locked Account Recovery via Signer Lists** (accounts) Add an lsfRecoveryList flag settable via SignerListSet on a dedicated signer-list slot, plus two AccountRoot SFields: sfRecoveryDelay (ledgers) and sf
- **Fixed-Term Amortizing Loans With Scheduled Claims** (lending) Add sfMaturityLedger, sfInstallmentAmount, sfInstallmentInterval, and sfNextInstallmentLedger to the loan object introduced by LendingProtocolV1_1, fi

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-22 11:27	09-22 11:27	xrpl-validator	`71fd2d5`	fix(ledger): F357 — OfferCreate judges "crossed" by the after-cross amounts, not by a dust output
09-22 14:11	09-22 14:11	xrpl-validator	`7c9fde0`	fix(ledger): F358 — OracleSet rebuilds the series as rippled does: unnamed pairs keep only their asset names, output in token-pair order
09-23 02:19	09-23 02:19	xrpl-validator	`c3c915c`	feat(node): receipt canary — the live shadow plants a 1-drop diff on demand and proves its compare still flags it
09-23 02:43	—	xrpl-validator	`12a043f`	fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0
09-23 02:53	—	xrpl-validator	`0e3f28b`	fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own
09-23 03:00	—	xrpl-validator	`28aec17`	fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)
09-23 03:06	09-23 03:06	xrpl-sdk	`c6fd55d`	fix(core): encode rippled's "1" spelling of noCurrency (finding 369)
09-23 03:06	—	xrpl-validator	`0a3dfb4`	fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19)

Committed	Merged into main	Repo	Commit	Subject
09-23 03:14	—	xrpl-validator	`dd90d91`	test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set
09-23 03:23	—	xrpl-validator	`fa5735e`	fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18)
09-23 03:30	—	xrpl-validator	`e2c806b`	fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)
09-23 03:33	—	xrpl-validator	`081001c`	fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)
09-23 03:49	—	xrpl-validator	`9010547`	fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)

Times in UTC. A commit appears on the day it was committed to main or merged into it.